Introduction

For many farmers in Pakistan, especially in Punjab and South Punjab, water is not just a farming need. It is the difference between profit and loss, crop success and crop failure, peace of mind and daily stress.

When a farmer thinks about irrigation, one question comes again and again:

Should I keep running my tube well on diesel, or should I convert it to solar?

The simple answer is this:

Solar tube wells usually save more money in the long run, especially for farmers who run their tube well regularly during the day. Diesel tube wells may cost less at the start, but their running cost is much higher because fuel, maintenance, engine oil, and repairs keep eating profit every month.

But the real answer is deeper than that.

Solar is not automatically right for every farmer. Diesel is not always wrong. The best choice depends on your land size, bore depth, water level, crop type, daily irrigation hours, pump size, budget, and whether you need water during the day or at night.

This guide explains the complete difference between solar and diesel tube wells in Pakistan. It will help you calculate which option saves more money, how long solar takes to recover its cost, when diesel still makes sense, and what mistakes to avoid before making a major investment.

Quick Answer: Which One Saves More Money?

In most regular farming cases, solar saves more money than diesel because sunlight has no daily fuel cost. Once the solar tube well system is installed properly, the farmer’s operating cost becomes very low compared to diesel.

However, diesel can still be useful when:

- The farmer needs water mostly at night
- The land is rented for short term

- The budget is very limited
- The tube well is used only occasionally
- The farmer needs backup during cloudy weather or emergency irrigation
- The bore has low water and solar investment may not be justified

Simple Comparison

Factor Solar Tube Well Diesel Tube Well Initial Cost High Lower Daily Running Cost Very low High Fuel Requirement No diesel required Diesel required daily Maintenance Low to medium Medium to high Best Use Regular daytime irrigation Backup or flexible timing Long-Term Saving Strong Weak Risk Wrong system sizing Diesel price increase Payback Usually possible with regular use No payback, only expense Best For Long-term farmers Short-term or emergency use

Final practical answer:If you run your tube well frequently and plan to farm for years, solar usually saves more money. If you use the tube well rarely or need mostly night irrigation, diesel may still have a role as backup.

What Is a Solar Tube Well?

A solar tube well uses solar panels to generate electricity and run a water pump. The pump pulls underground water from the bore and delivers it to the field, water channel, tank, pond, drip system, sprinkler, or irrigation pipeline.

A complete solar tube well system usually includes:

Component Purpose Solar panels Generate electricity from sunlight VFD / solar pump controller Controls pump operation Pump and motor Pull water from underground Structure Holds the solar panels DC/AC cables Transfer power safely Protection system Protects pump and controller Pipes and fittings Carry water to field Installation work Ensures system runs properly

A solar tube well is not just a panel setup. It is a complete water project. If one part is weak, the whole system can underperform.

What Is a Diesel Tube Well?

A diesel tube well uses a diesel engine to run the pump. It has been used for decades in Pakistan because it is flexible and does not depend on grid electricity.

A diesel setup usually includes:

Component Purpose Diesel engine Provides mechanical power Pump Pulls water from bore Fuel tank Stores diesel Belt/shaft/coupling Transfers engine power Engine oil and filters Required for maintenance Delivery pipe Carries water to field

Diesel tube wells are simple and familiar, but their biggest problem is running cost. Every hour of operation burns fuel. Every season needs maintenance. Every diesel price increase directly affects the farmer.

The Real Difference: One Has Fixed Cost, One Has Daily Cost

This is the most important point.

A solar tube well has a high one-time cost but very low daily cost.

A diesel tube well has a lower starting cost but high daily cost.

Cost Behavior

Cost Type Solar Tube Well Diesel Tube Well One-time investment High Low to medium Daily cost Very low High Monthly cost Low High Seasonal cost Predictable Unpredictable Long-term cost Decreases over time Keeps increasing Cost control Better Weak

This is why solar feels expensive on day one, but diesel becomes expensive every month.

Solar vs Diesel Tube Well: Full Comparison

Feature Solar Tube Well Diesel Tube Well Energy Source Sunlight Diesel fuel Running Cost Very low High Fuel Availability Not needed Must be purchased Maintenance Panels cleaning, VFD/pump checks Engine oil, filters, belts, fuel system, repairs Noise Low High Smoke/Pollution No smoke during operation Smoke and emissions Daytime Use Excellent Good Night Use Needs grid/battery/backup Good Long-Term Economy

Strong Weak Initial Investment High Lower Ease of Start Automatic/controlled Manual/mechanical Best For Regular irrigation Backup and flexible timing Risk Poor design or theft Fuel price and engine breakdown

Diesel Tube Well Cost: The Expense Farmers Feel Every Day

Diesel cost is not just the price of fuel. It includes:

- Diesel consumption per hour
- Engine oil
- Filters
- Belt/coupling maintenance
- Mechanic charges
- Engine wear
- Fuel transport
- Time wasted arranging diesel
- Breakdown during crop season
- Unplanned repair cost

Many farmers only calculate diesel fuel cost. That gives an incomplete picture.

Diesel Running Cost Formula

Use this simple formula:

Monthly diesel cost = litres per hour × diesel price × running hours per day × working days per month

Example:

If a diesel tube well consumes 3 litres per hour, diesel price is PKR 300 per litre, and the pump runs 5 hours per day for 25 days, then:

3 × 300 × 5 × 25 = PKR 112,500 per month

That is only fuel cost. It does not include engine oil, repair, transport, and maintenance.

Estimated Diesel Cost by Pump Usage

These are practical examples. Actual cost can change depending on engine condition, pump size, water depth, diesel price, and running hours.

Usage Level Diesel Consumption Daily Running Monthly Fuel Cost Estimate Light Use 1.5 litres/hour 3 hours/day PKR 30,000 – 45,000 Medium Use 2.5 litres/hour 4 hours/day PKR 70,000 – 90,000 Heavy Use 3.5 litres/hour 5 hours/day PKR 125,000 – 150,000 Very Heavy Use 5 litres/hour 6 hours/day PKR 225,000 – 270,000

For a farmer running diesel regularly, this becomes a serious yearly burden.

Yearly Diesel Cost: The Number That Changes the Decision

A farmer may not feel the full cost in one day. But when you calculate diesel cost yearly, the result is serious.

Monthly Diesel Cost Yearly Diesel Cost PKR 50,000 PKR 600,000 PKR 75,000 PKR 900,000 PKR 100,000 PKR 1,200,000 PKR 150,000 PKR 1,800,000 PKR 200,000 PKR 2,400,000 PKR 250,000 PKR 3,000,000

This is why solar becomes attractive. A farmer who spends PKR 1.2 million to PKR 2.4 million per year on diesel is not just buying fuel. He is slowly paying for a solar system without owning one.

Solar Tube Well Cost: The Big Investment That Pays Back

A solar tube well costs more upfront because you are buying the power source in advance. The panels, structure, VFD, pump compatibility, wiring, and installation all come at the start.

Typical system prices can vary like this:

Pump Size Solar Capacity Estimated Complete System Cost 5HP 5kW – 7kW PKR 8 lakh

– 12 lakh 10HP 10kW – 14kW PKR 15 lakh – 22 lakh 15HP 15kW – 20kW PKR 22 lakh – 30 lakh 20HP 20kW – 28kW PKR 28 lakh – 40 lakh 30HP+ 30kW+ PKR 40 lakh – 60 lakh+

These ranges are not fixed. A deeper bore, longer cable, stronger structure, better VFD, better pump, and better protection system can increase the cost.

But unlike diesel, solar does not require daily fuel.

Solar Payback Period: The Most Important Calculation

The payback period tells you how long it takes for solar to recover its cost through diesel savings.

Formula

Solar payback period = total solar system cost ÷ monthly diesel saving

Example:

If your solar system costs PKR 1,800,000 and you are saving PKR 100,000 per month in diesel, then:

1,800,000 ÷ 100,000 = 18 months

That means the system can recover its cost in about one and a half years.

After that, the farmer continues saving money for years.

Payback Example: 5HP Tube Well

Let's say a farmer has a small to medium farm and runs a 5HP diesel tube well.

Diesel Case

Item Estimate Diesel consumption 1.5 – 2.5 litres/hour Daily use 4 hours Monthly use 25 days Diesel price assumption PKR 300/litre Monthly fuel cost PKR 45,000 – 75,000 Yearly fuel cost PKR 540,000 – 900,000

Solar Case

Item Estimate Solar system size 5kW – 7kW Approx. system cost PKR 8 lakh – 12 lakh Monthly diesel saving PKR 45,000 – 75,000 Approx. payback 12 – 24 months

Result

For a farmer using the tube well regularly, solar can make strong financial sense. If the tube well is used only occasionally, payback will take longer.

Payback Example: 10HP Tube Well

A 10HP tube well is common for serious agricultural use.

Diesel Case

Item Estimate Diesel consumption 2.5 – 4 litres/hour Daily use 5 hours Monthly use 25 days Diesel price assumption PKR 300/litre Monthly fuel cost PKR 93,750 – 150,000 Yearly fuel cost PKR 1,125,000 – 1,800,000

Solar Case

Item Estimate Solar system size 10kW – 14kW Approx. system cost PKR 15 lakh – 22 lakh Monthly diesel saving PKR 93,750 – 150,000 Approx. payback 12 – 24 months

Result

For a 10HP regular-use tube well, solar usually wins strongly. The diesel cost is high enough that the solar system can recover its investment in a practical time.

Payback Example: 15HP Tube Well

A 15HP system is used for larger farms or higher water demand.

Diesel Case

Item Estimate Diesel consumption 4 – 6 litres/hour Daily use 5 hours Monthly use 25 days Diesel price assumption PKR 300/litre Monthly fuel cost PKR 150,000 – 225,000 Yearly fuel cost PKR 1,800,000 – 2,700,000

Solar Case

Item Estimate Solar system size 15kW – 20kW Approx. system cost PKR 22 lakh – 30 lakh Monthly diesel saving PKR 150,000 – 225,000 Approx. payback 12 – 20 months

Result

For heavy use, solar can save a large amount of money. But the system must be designed professionally because a weak 15HP solar system can create major performance problems.

Solar Saves More Only When the System Is Designed Correctly

Solar does not save money if the system is poorly designed.

A wrong solar tube well system can cause:

- Low water discharge
- Pump tripping
- VFD faults
- Weak morning/evening performance
- Motor heating
- Cable loss
- Panel underperformance
- Crop irrigation delays
- Extra repair cost

The saving comes from correct design.

A proper solar tube well design must consider:

Factor Why It Matters Bore depth Decides pump head Static water level Shows water level before pumping Dynamic water level Shows water level during pumping Pump HP Decides power requirement Water flow Decides irrigation capacity Solar kW Decides available energy VFD size Controls motor safely Cable length Affects voltage drop

Structure strength Protects panels Crop type Decides water demand Irrigation method Decides pressure and flow

The cheapest quote is often not the cheapest long-term solution.

Hidden Costs of Diesel Tube Wells

Diesel looks cheaper because the engine may already be installed. But hidden costs continue every season.

1. Fuel Price Risk

Diesel prices change. A farmer cannot control this. If diesel increases, irrigation cost increases immediately.

2. Engine Maintenance

Diesel engines need regular maintenance:

- Engine oil
- Filters
- Belt adjustment
- Injector service
- Pump alignment
- Cooling system
- Mechanical repairs

3. Breakdown During Critical Crop Stage

If the engine fails during sowing, flowering, or fruit development, the cost is not just repair. The crop can suffer.

4. Time and Labour

Someone must arrange fuel, start the engine, monitor it, and manage breakdowns.

5. Lower Predictability

With diesel, the farmer keeps thinking about the next fuel purchase. With solar, the main cost is already paid.

Hidden Costs of Solar Tube Wells

Solar also has hidden costs if not planned properly.

1. Weak Structure

A poor structure can bend, rust, or fail in strong wind. Good structure costs more but protects the panels.

2. Poor VFD

A low-quality or wrongly sized VFD can damage pump performance.

3. Undersized Solar Panels

If panels are fewer than required, the pump may not perform properly.

4. Theft Risk

Solar panels installed in open fields need security planning.

5. Cleaning and Maintenance

Dust reduces panel performance. Panels need cleaning, especially in agricultural areas.

6. Water Mismanagement

Because solar running cost is low, farmers may over-pump water. This can affect long-term water availability.

When Solar Tube Well Saves the Most Money

Solar saves the most money when:

- You run the tube well regularly

- Your diesel cost is high
- You use water mostly during the day
- Your bore has good water yield
- Your pump size is 5HP or above
- You own the land long-term
- Your crops need regular irrigation
- You can protect the solar structure
- You want predictable farming cost
- You have proper system design

Solar is not just about saving fuel. It is about controlling your farming cost.

When Diesel Tube Well Still Makes Sense

Diesel can still be useful in some situations.

Diesel May Be Better If:

- You use the tube well only a few times per month
- You need water mostly at night
- You cannot invest in solar right now
- You are farming rented land for a short period
- Your bore water is uncertain
- Your land has serious theft risk
- You need emergency backup
- Your water requirement is very irregular

In these cases, a diesel engine may remain useful as backup.

Best Practical Option: Solar + Diesel Backup

For many farmers, the best solution is not solar vs diesel. It is:

Solar as the main system + diesel as backup.

This gives the farmer both savings and flexibility.

Use Best Source Regular daytime irrigation Solar Night emergency irrigation Diesel Cloudy weather backup Diesel Main cost saving Solar Crop protection backup Diesel

This hybrid thinking is practical because agriculture cannot depend on one condition only. Crops need water at the right time.

Solar vs Diesel for Different Farmer Types

Farmer Type Better Option Reason Small farmer using tube well rarely Diesel or small solar Payback may be slow Small farmer using daily irrigation Solar Regular use improves savings Medium farmer Solar Strong saving potential Large farmer Solar + backup High diesel cost makes solar attractive Orchard owner Solar Regular watering schedule fits solar Rice grower Solar with water control Water use must be managed carefully Vegetable farmer Solar Frequent irrigation benefits from low running cost Rented land farmer Depends Solar only if long-term contract exists Farmhouse owner Solar Low running cost and easy operation Commercial farm Engineered solar system High use requires professional design

Environmental and Water Management Difference

Solar is cleaner than diesel because it avoids smoke, noise, and daily fuel burning. But solar also creates a new responsibility: water discipline.

Because solar water is cheap to pump, some farmers start pumping more than needed. That can reduce the underground water level over time.

A smart farmer should use solar with proper water management:

- Avoid over-irrigation
- Use water according to crop stage
- Repair water channels
- Use pipelines where possible
- Consider drip, sprinkler, or center pivot irrigation
- Monitor bore water output
- Check water quality
- Plan irrigation schedule
- Avoid wasting water in open channels

The real future is not only solar pumping. The real future is solar pumping plus smart water management.

Cost Comparison Over 5 Years

This table shows why solar usually wins long term.

Example: Medium Farmer

Cost Area Diesel Tube Well Solar Tube Well Initial setup Lower Higher Monthly fuel High Almost zero Maintenance High Low to medium 1-year cost High High first year 3-year cost Very high Usually recovered 5-year cost Extremely high Much lower Cost control Weak Strong

5-Year Cost Illustration

Assume a farmer spends PKR 100,000 per month on diesel.

Year Diesel Cost Solar Cost Position Year 1 PKR 1,200,000 Solar investment recovering Year 2 PKR 2,400,000 total Solar may already be recovered Year 3 PKR 3,600,000 total Solar gives strong saving Year 4 PKR 4,800,000 total Diesel cost keeps rising Year 5 PKR 6,000,000 total Solar advantage becomes clear

If a solar system costs PKR 18 lakh and replaces PKR 100,000 monthly diesel cost, the farmer may recover the system in around 18 months. After that, the saving becomes real profit support.

Which One Is Better for Crop Profit?

The better system is the one that gives water at the right time with the lowest long-term cost.

Diesel Impact on Profit

Diesel reduces profit because every irrigation cycle has a fuel cost. If crop prices fall or input costs rise, diesel becomes even more painful.

Solar Impact on Profit

Solar improves profit because the farmer can irrigate during daylight without worrying about daily fuel cost. This can improve crop timing and reduce stress.

But solar can hurt long-term farming if the farmer wastes water. More pumping is not always better. Correct pumping is better.

Decision Table: Solar or Diesel?

Use this table before making a decision.

Question If Your Answer Is Yes Better Option Do you run tube well regularly? Yes Solar Do you spend heavily on diesel monthly? Yes Solar Do you need mostly daytime irrigation? Yes Solar Do you own the land long-term? Yes Solar Is your bore water reliable? Yes Solar Do you need mostly night water? Yes Diesel backup Do you use tube well rarely? Yes Diesel or small solar Is your bore water uncertain? Yes Survey first Is theft risk high? Yes Security planning needed Is budget very limited? Yes Phased approach

Before Converting Diesel Tube Well to Solar

Do not convert blindly. First, inspect the current system.

Conversion Checklist

Item Check Before Solar Bore depth Required Water level Required Existing pump HP Required Pump condition Required Motor condition Required Current water flow Required Diesel consumption Required Daily running hours Required Crop water need Required Cable and pipe route Required Solar panel space Required Theft/security risk Required

A good conversion starts with facts, not guesswork.

Common Mistakes Farmers Make

Mistake 1: Comparing Only Initial Price

A diesel engine may look cheaper today. But the real cost is monthly fuel.

Mistake 2: Buying Cheap Solar Without Design

Cheap solar can fail to deliver water. A wrong system wastes money.

Mistake 3: Ignoring Bore Condition

If the bore is weak, solar will not solve the water problem.

Mistake 4: Using Wrong Pump

Pump selection must match head, flow, bore size, and water level.

Mistake 5: Ignoring VFD Quality

The VFD controls the motor. A bad VFD can damage performance.

Mistake 6: No Backup Plan

Even solar farms may need backup for night or emergency irrigation.

Mistake 7: No Water Management

Cheap water does not mean unlimited water. Over-pumping can damage future farming.

What Should a Professional Solar Tube Well Quotation Include?

A serious quotation should include more than one line.

Quotation Detail Why It Matters Pump HP Shows system capacity Solar kW Shows power generation Panel brand/wattage Affects performance VFD brand/rating Affects motor safety Pump type Affects water output Cable size Affects safety and efficiency Structure design Affects durability Protection system Prevents damage Installation scope Avoids hidden charges Warranty Shows reliability After-sales support Important after installation Site assumptions Prevents wrong expectations

If the quotation only says “10HP solar system package,” ask for details.

Practical Recommendation

Choose Solar If:

- You are tired of diesel cost
- You run the tube well regularly
- You want long-term savings
- Your bore has reliable water
- You use irrigation mostly in daytime
- You plan to farm long term
- You want predictable operating cost

Keep Diesel Backup If:

- You need emergency night irrigation
- You grow sensitive crops
- Weather conditions affect irrigation timing
- Your crop cannot wait for sunlight
- You want extra security

Do Not Install Solar Yet If:

- Your bore is failing
- Water flow is very low
- You do not know your water depth
- You cannot secure the site
- You are leaving the land soon
- Your budget only allows poor-quality equipment

Best Strategy for Pakistani Farmers

The best strategy is not emotional. It is mathematical.

Before deciding, calculate:

- How many litres of diesel you use per hour
- How many hours you run the tube well per day
- How many days you run it per month
- Your monthly diesel cost
- Your yearly diesel cost
- Solar system cost for your pump size
- Expected payback period
- Bore condition
- Crop water requirement
- Backup need

If solar payback is under 2 to 3 years and you plan to farm long-term, solar is usually a strong decision.

Simple Farmer Calculation Sheet

Use this simple sheet.

Question Your Answer Current pump HP ____ Diesel used per hour ____ litres Diesel price per litre PKR ____ Daily running hours ____ hours Monthly running days ____ days Monthly diesel cost PKR ____ Yearly diesel cost PKR ____ Solar system quotation PKR ____ Expected monthly saving PKR ____ Payback period ____ months

Formula

Monthly Diesel Cost = Diesel Litres per Hour × Diesel Price × Daily Hours × Monthly Days

Payback Period = Solar System Cost ÷ Monthly Diesel Saving

This simple calculation can save a farmer from making a wrong investment.

Final Verdict: Which One Saves More Money?

For most regular-use agricultural tube wells in Pakistan:

Solar saves more money than diesel.

Diesel may look easier because the investment is smaller at the beginning. But diesel keeps taking money every day. Solar requires a bigger investment at the start, but it reduces the running cost dramatically.

The farmer who uses water regularly and owns land long-term should seriously consider solar. The farmer who uses water rarely or needs mostly night irrigation may keep diesel or use diesel as backup.

The smartest solution for many farmers is:

Solar for daily irrigation + diesel as emergency backup + proper water management.

This gives cost saving, reliability, and control.

Final CTA

Planning to Convert Your Diesel Tube Well to Solar?

FJ Group can help you calculate whether solar is actually profitable for your land.

Our team can review your current diesel usage, pump size, bore depth, water requirement, crop type, and daily irrigation pattern. Then we can guide you with a practical solar tube well plan that is designed for performance, not just price.

Whether you need:

- Diesel tube well to solar conversion
- New solar tube well installation
- Bore and water condition assessment
- Pump and VFD selection
- Solar water pumping system
- Farm irrigation planning
- Water management solution

FJ Group can help you make the right decision before you invest.

Contact FJ Group today for solar tube well consultation and project quotation.

Website: www.fjgroup.pkPhone: 03 111 777 286

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